# SUHAS KOTA

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## SUMMARY

B.Tech Computer Science student with strong foundations in Data Structures and Algorithms and hands on experience in backend development and machine learning. Built REST APIs, semantic search systems, and predictive models through academic projects and competitive programming. Focused on developing efficient, scalable, and well-structured software solutions.

## PROJECTS

**Rooftop Rainwater Harvesting & Recharge WebApp** GitHub

- Developed Flask based backend to assess rooftop rainwater harvesting and groundwater recharge potential.
- Built groundwater level forecasting models with scikit-learn, trained on 10+ years of historical rainfall data.
- Built data preprocessing pipeline using Pandas/NumPy and stored structured data in PostgreSQL.
- Implemented algorithm to compute optimal pit/tank dimensions based on rooftop area and rainfall stats.
- Designed and tested REST APIs using Postman; managed version control with Git.

**Scientific Literature Explorer** GitHub | Demo

- Built embedding based semantic search system for academic research papers.
- Implemented PDF extraction and contextual chunking pipeline for improved information retrieval.
- Generated embeddings using Sentence Transformers and applied cosine similarity for query matching.
- In Developed citation graph visualization using NetworkX and similarity-based paper recommendations.
- Deployed interactive application using Streamlit.

## EXPERIENCE

**Smart India Hackathon (SIH) Participation**

- Integrated machine learning models with scikit-learn to predict groundwater levels using historical datasets.
- Designed backend logic to store user input and forecast data in PostgreSQL; handled data manipulation using Pandas and NumPy.
- Developed and tested REST APIs for interaction between frontend and model logic using Postman.
- Presented the solution during the internal round of SIH, focusing on usability for households to urban planning.

**Competitive Programming**

- Solved 300+ algorithmic problems on LeetCode under strict time constraints.
- Applied core data structures and algorithms to build time and space optimized solutions.
- Strong in arrays, trees, graphs, recursion, and dynamic programming.
- Utilized Python (including csv and re) for efficient parsing and problem solving.

## EDUCATION

**Bachelor of Technology (B.Tech) in Computer Science & Engineering** — **Aug 2023 – Present**
GITAM Deemed University, Visakhapatnam – **CGPA:** 9.28

**High School** — **Apr 2011 – May 2021**
Delhi Public School, Visakhapatnam – **Percentage:** 88%

## SKILLS

- **Languages:** Python, Java, C, Kotlin
- **Computer Science Fundamentals:** Data Structures, Algorithms, OOPs, DBMS, OS, Complexity Analysis
- **Backend & Database:** REST APIs, Flask, SQL
- **Machine Learning & Data Science:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib
- **Tools & Platform:** Git, GitHub, Streamlit, Postman, Tailscale, VS Code, Google Colab
- **Soft Skills:** Communication, Problem Solving, Adaptable in Challenges, Team Leadership and Collaboration

## CERTIFICATIONS

Android-mobile-Application-Development

Kotlin

Hackathons